



Algebra 1 exam

Note /20

First and last name	Serial number	Group

Exercise (01): (09 points). Let $G = \mathbb{R} - \{-3\}$. We define on G an internal composition law denoted $*$ by:

$$\forall x, y \in G, \quad x * y = xy + 3(x + y) + 6.$$

1. Show that $(G, *)$ is an abelian group.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. A vertical line runs down the center of the page, creating two equal-width columns. The paper appears to be from a notebook or a standard writing template. There are no markings, text, or drawings on the page.

[illegible]

P_1 : "The maps of the functions f and g intersect at a single point in the interval $[-\pi, \pi]$."

- P_1 :
- $\overline{P_1}$:
- Is the proposition P_1 true for the applications $f(x) = \frac{1}{2} + \frac{1}{2}\cos^2(x)$ and $g(x) = -\cos(x)$?

Exercise (03): (02points). Prove the following proposition:

$$\forall (a, b) \in \mathbb{R}^2 : a^2 + b^2 = 1 \Rightarrow |a + b| \leq \sqrt{2}.$$

Exercise (04): (06points). Let f be an application from $\mathbb{R}$ to $\mathbb{R}$ defined by:

$$f(x) = \ln\left(\frac{e^x}{1 + e^x}\right).$$

1. Determine the following sets:

$$f(\{-\ln(3); 0\}), \quad f^{-1}(\{1; -2\}) \text{ and } f^{-1}([-1, 1[).$$

4. Show that f is bijective from $\mathbb{R}$ to $] -\infty, 0[$ and give the inverse application f^{-1} .